IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
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- Methodology
- Results
- Conclusion
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Executive Summary

- This project aims to predict the success of SpaceX Falcon 9 first-stage landings. SpaceX offers launches at approximately $62 million, significantly lower than the $165 million charged by other providers, largely due to its ability to reuse the first-stage booster.

- Data was collected through APIs and web scraping, then cleaned and structured using data wrangling techniques. Exploratory Data Analysis (EDA) and visualization were conducted to identify key patterns and trends. After feature engineering and preprocessing, four machine learning models were trained to predict landing outcomes.

- All models achieved similar performance, with an average accuracy of about 83%.

Introduction

- We are part of the data science team at SpaceY, analyzing publicly available SpaceX data to predict Falcon 9 first-stage landing success. Since reusability significantly reduces launch costs, accurate predictions provide a competitive advantage in bidding strategies.

- Data was collected via the SpaceX API using HTTP requests and BeautifulSoup, then cleaned and structured with pandas and SQLite. Exploratory analysis and visualizations (Pandas, Seaborn) identified key factors influencing landing outcomes.

- We also used Folium for launch site mapping and Dash to build an interactive dashboard.

- The problem was treated as a binary classification task using Logistic Regression, SVM, Decision Tree, and KNN with GridSearchCV. While the baseline success rate is 66.67%, our models achieved approximately 83% accuracy.

Section 1

Methodology

Methodology

- **Objective:** Predict Falcon 9 first-stage landing success using publicly available SpaceX data.
- **Data Collection:** Extracted data via API and web scraping (HTTP, BeautifulSoup), structured with pandas.
- **Data Processing:** Cleaned, transformed, and queried data using pandas and SQL.
- **EDA & Visualization:** Identified key factors (payload, orbit, flight number) using SQL, Matplotlib, and Seaborn.
- **Interactive Analytics:** Developed launch site maps (Folium) and dashboards (Plotly Dash).
- **Predictive Modeling:** Built and tuned Logistic Regression, SVM, Decision Tree, and KNN models with GridSearchCV.
- **Results:** Achieved ~83% accuracy (baseline success rate: 66.67%).

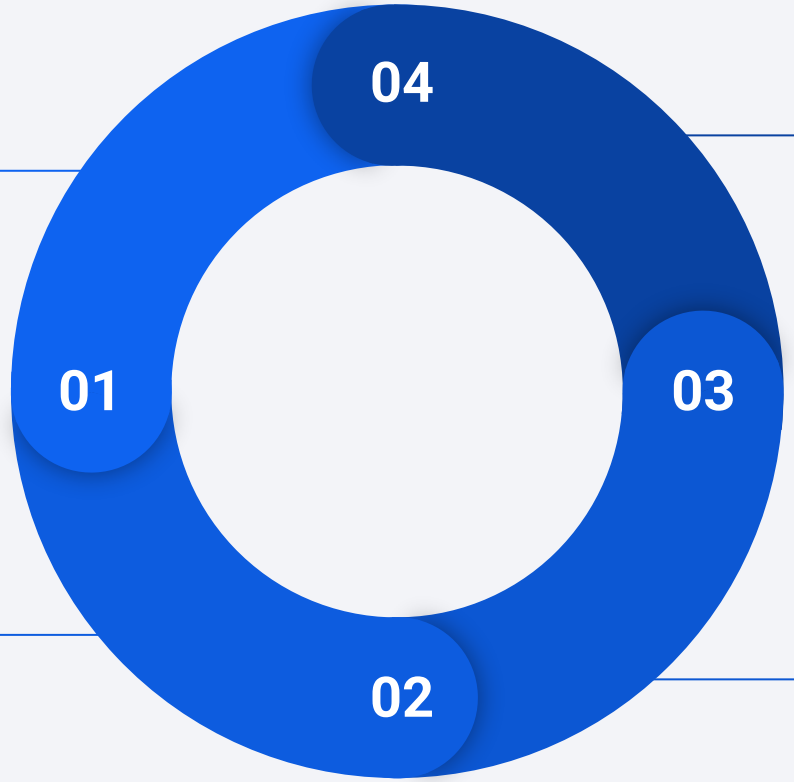
Data Collection

API request

Data was collected from the **SpaceX public API** using Python HTTP requests, which returned launch information in **JSON format**..

Data Frame

The JSON data was converted into a **pandas DataFrame** to organize and structure the information for analysis.



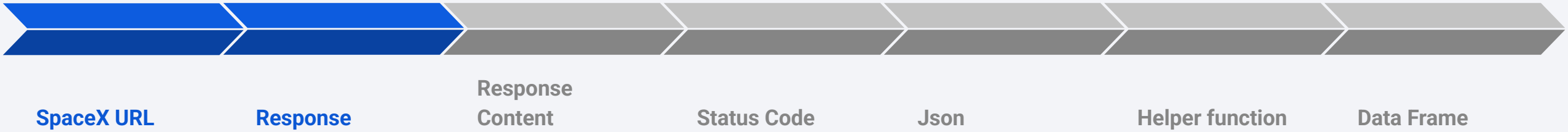
Save/ Export Cleaned data

The cleaned dataset was saved and exported for further **exploratory analysis and machine learning modeling**.

Data Cleaning

Irrelevant data (Falcon 1 launches) was removed, and missing values were handled to improve the dataset quality.

Data Collection – SpaceX API



[Github SpaceX API calls notebook]

Data Collection - Scraping

Static URL | **Request get Text** | Beautiful Soup | Tables | Column names | Helper function | Data Frame

Github SpaceX Webscrapping notebook

Data Wrangling

- The cleaned dataset was first imported for analysis. We began by examining the percentage of missing values in each column.
- The **LaunchingPad** column was the only field containing missing values, representing cases where no launching pad was used.
- We then reviewed the **data types** of all variables. Four different data types were identified: **int64, object, float64, and bool**.
- Additional exploratory analysis was performed, including a **value count of the LaunchSite variable**.
- The results showed that **Cape Canaveral Space Launch Complex 40** and **VAFB SLC-4E** recorded the highest number of launches, with **55 occurrences**.

[Github Data Wrangling notebook]

EDA with Data Visualization

- We used **Seaborn** and **Matplotlib** to perform visual analysis of the dataset and better understand key patterns.
- We explored the **relationship between payload mass and launch sites** to observe how payload distribution varies across different launch facilities.
- We also examined the **connection between orbit types and launch success**, allowing us to identify which orbit categories were associated with higher success rates.
- Finally, we analyzed the **evolution of rocket launch success rates over time (2010–2020)** in order to observe long-term trends in launch performance.

[Github EDA notebook]

EDA with SQL

- Identification of **unique launch sites**
- Retrieval of **launch sites starting with “CCA”**
- Calculation of **total NASA CRS payload mass**
- Analysis of **average payload mass for F9 v1.1 boosters**
- Identification of **first successful ground pad landing**
- Analysis of **drone ship landings with payload 4000–6000 kg**
- Comparison of **successful vs failed missions**
- Identification of **boosters with maximum payload capacity**
- Analysis of **drone ship landing failures in 2015**
- Landing outcome analysis **(2010–2017)**

Github EDA with SQL notebook

Build an Interactive Map with Folium

- We created an **interactive map using Folium** to visualize the geographical locations of the launch sites.
- **Markers** were added to identify and pin the exact locations of different launch facilities, such as the **NASA JSC space station launch site**.
- **Circles** were used to highlight specific areas and emphasize important locations, accompanied by descriptive labels.
- We also added **lines to illustrate the proximity and spatial relationships between launch sites**, providing a clearer understanding of their geographical distribution.

[Github Interactive Map with Folium notebook]

Build a Dashboard with Plotly Dash

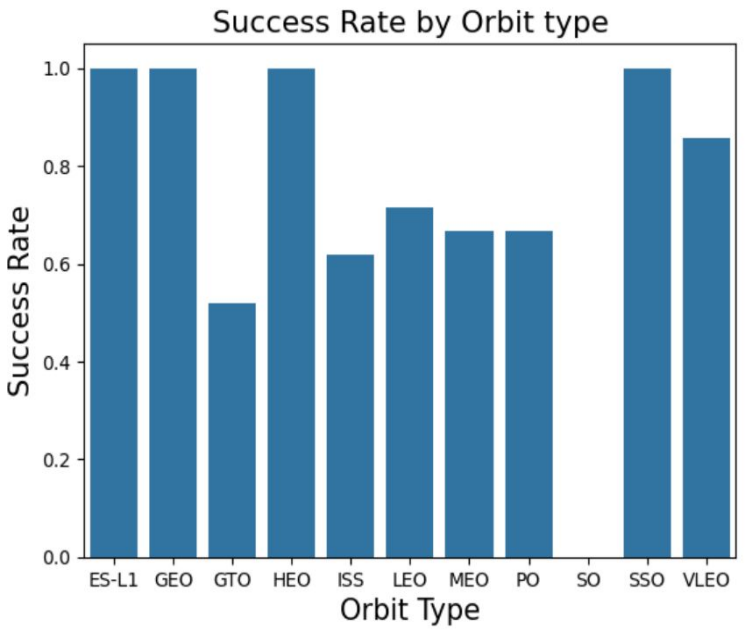
- We developed an **interactive dashboard using Plotly Dash** to enable dynamic data exploration.
- A **dropdown menu** was included to allow users to filter and select specific launch sites.
- A **range slider** was added to adjust the payload mass range for analysis.
- These interactive controls allow users to **explore the data more effectively and observe how different factors influence launch outcomes** through the visualizations.

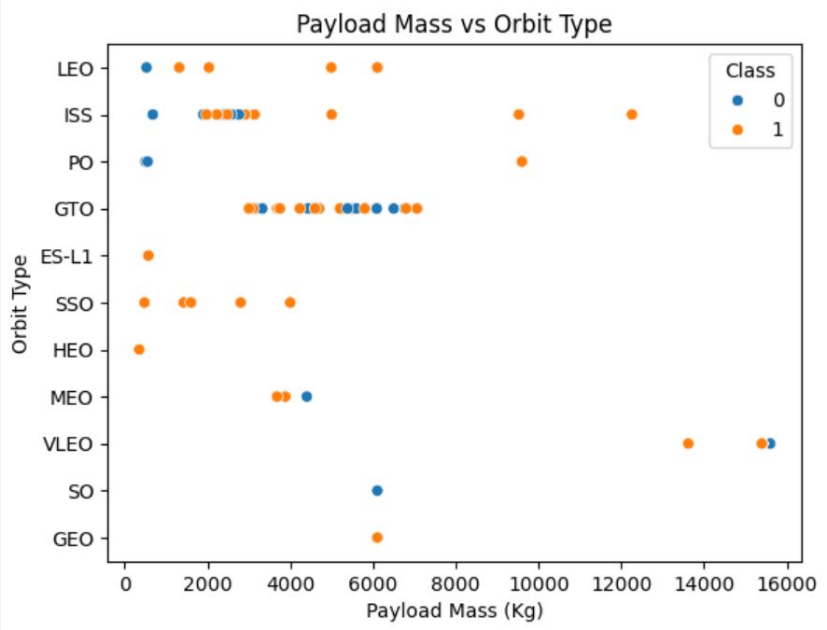
Predictive Analysis (Classification)

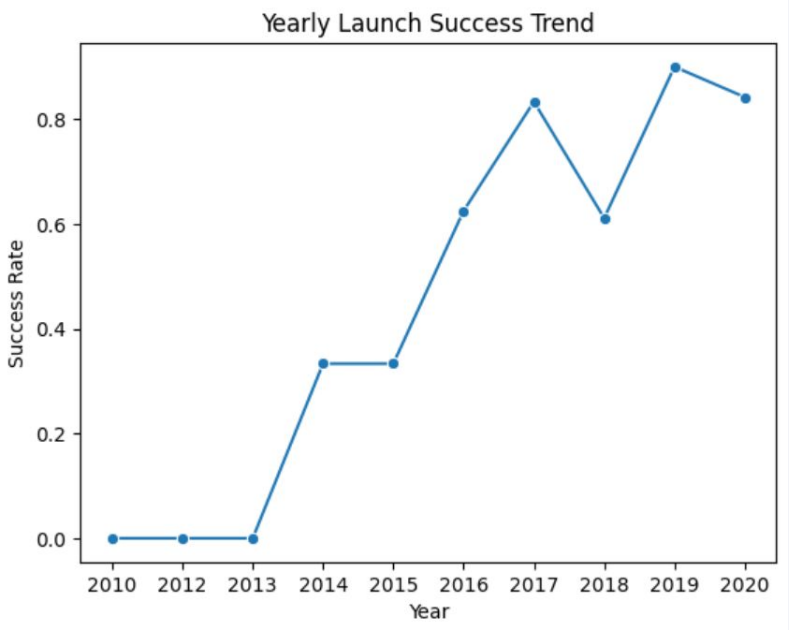
- We imported the cleaned dataset and separated the **features (X)** from the **target variable (Y)**, where the target represents the landing outcome.
- The feature variables were then **standardized using StandardScaler** to ensure that all variables were on a similar scale. After preprocessing, the dataset was divided into **training and testing sets**, with **20% of the data reserved for testing**.
- Several machine learning algorithms were trained, including **Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)**. Hyperparameter tuning was performed using **GridSearchCV with 10-fold cross-validation** to determine the optimal model configurations.
- Each model was trained using the training dataset and then evaluated on the **unseen test dataset** to measure its predictive performance.
- Among the tested models, all models produced similar **test accuracies of around 83%**.

Github Predictive analysis notebook

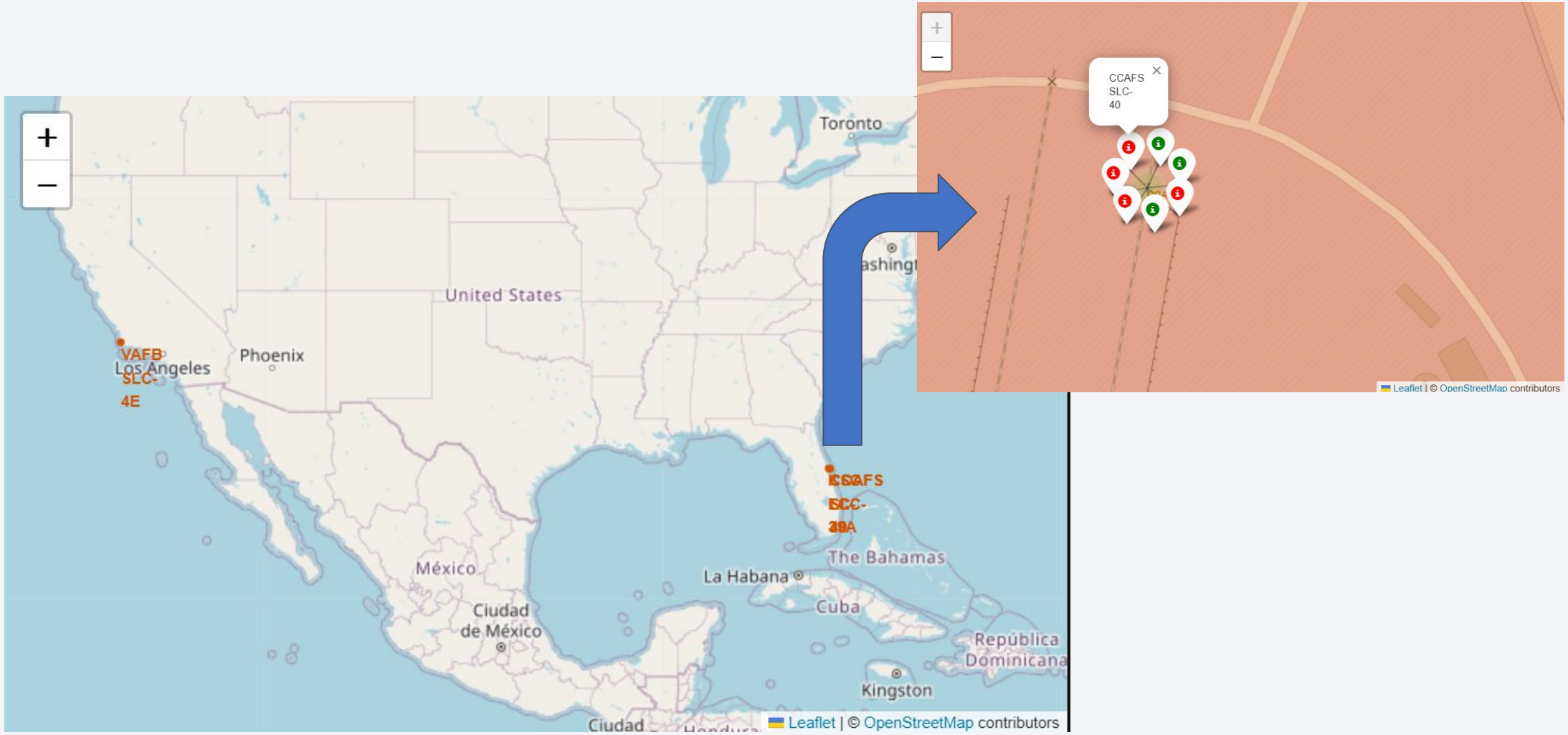
Results


Success Rate by Orbit type
Success Rate
1.0
0.8
0.6
0.4
0.2
0.0
ES-L1
GEO
GTO
HEO
ISS
LEO
MEO
PO
SO
SSO
VLEO
Orbit Type


Payload Mass vs Orbit Type
Orbit Type
LEO
ISS
PO
GTO
ES-L1
SSO
HEO
MEO
VLEO
SO
GEO
Class
0
1
0
2000
4000
6000
8000
10000
12000
14000
16000
Payload Mass (Kg)


Yearly Launch Success Trend
Success Rate
0.8
0.6
0.4
0.2
0.0
2010
2012
2013
2014
2015
2016
2017
2018
2019
2020
Year

Results


Toronto
United States
Phoenix
VAFB
Los Angeles
SLC-
4E
CCAFS
SLC-
40
The Bahamas
La Habana
Cuba
México
Ciudad
de México
República
Dominicana
Kingston
Leaflet | © OpenStreetMap contributors
Leaflet | © OpenStreetMap contributors

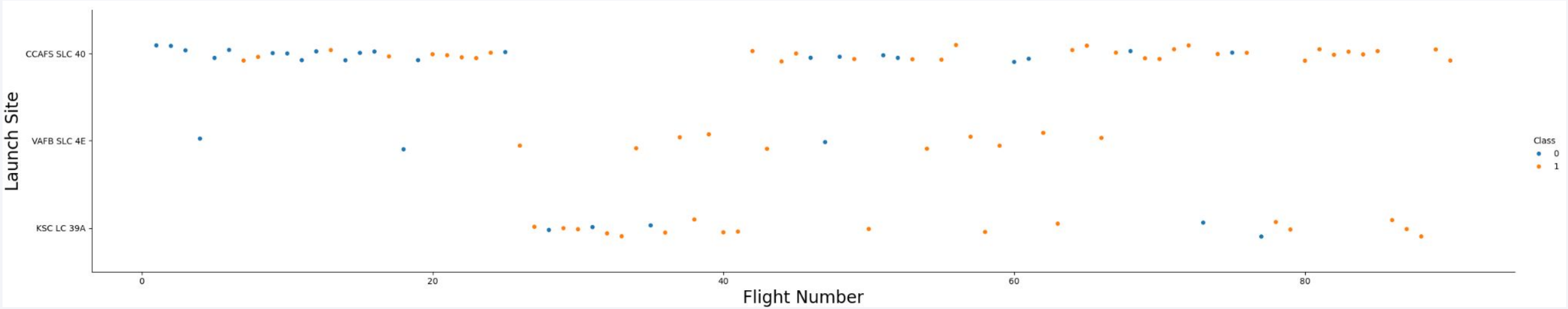
Results

Model	Best Hyperparameters (GridSearchCV)	Cross-Validation Accuracy	Test Accuracy
Logistic Regression	C = 0.01, penalty = L2, solver = lbfgs	0.846	0.833
Support Vector Machine (SVM)	Kernel selected through GridSearch optimization	—	0.833
Decision Tree	criterion = entropy, max_depth = 6, max_features = sqrt	**0.914 (Best CV Score)**	0.833
K-Nearest Neighbors (KNN)	n_neighbors = 10, p = 1, algorithm = auto	0.848	0.833

Section 2

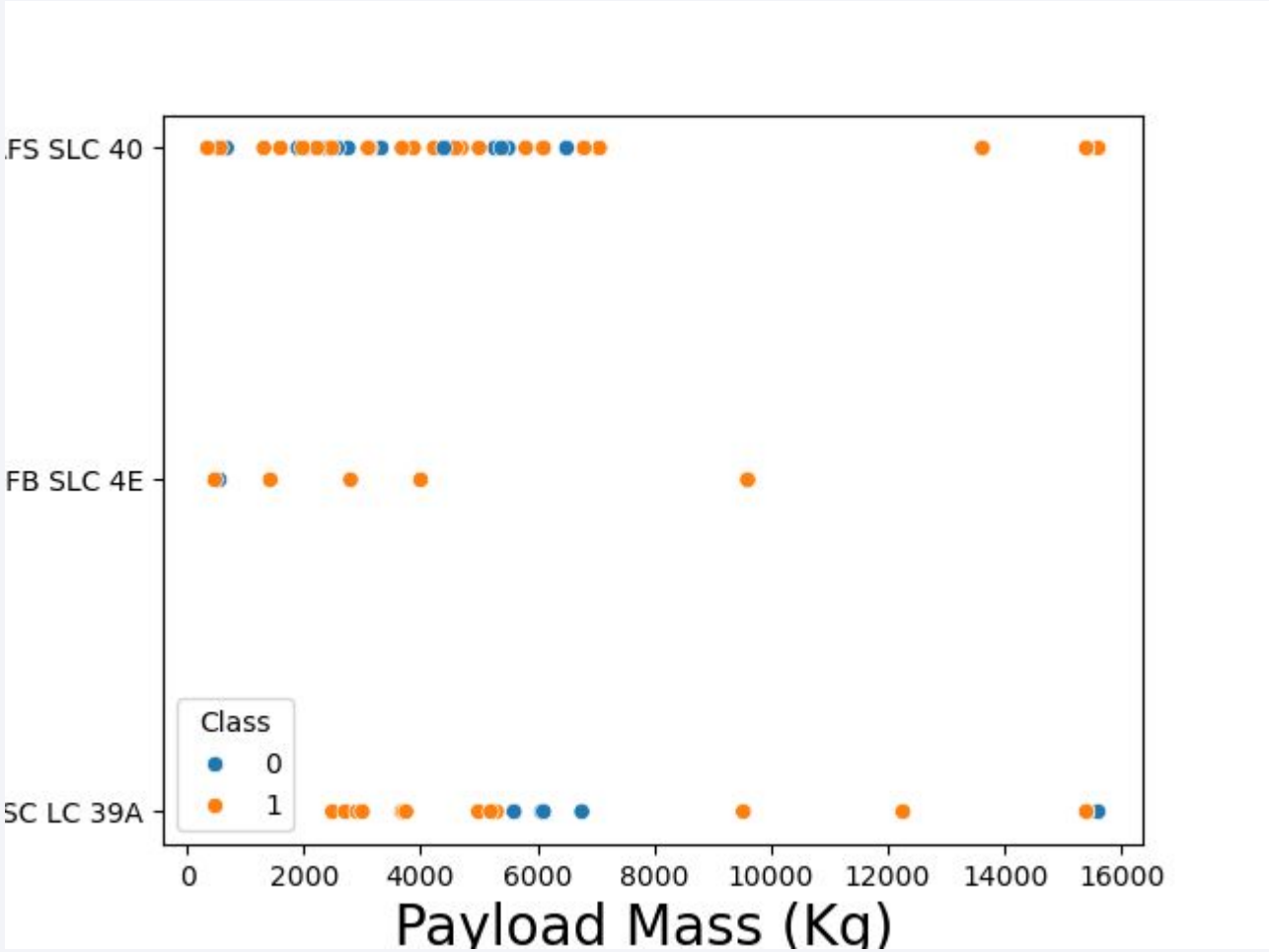
Insights drawn from EDA

Flight Number vs. Launch Site



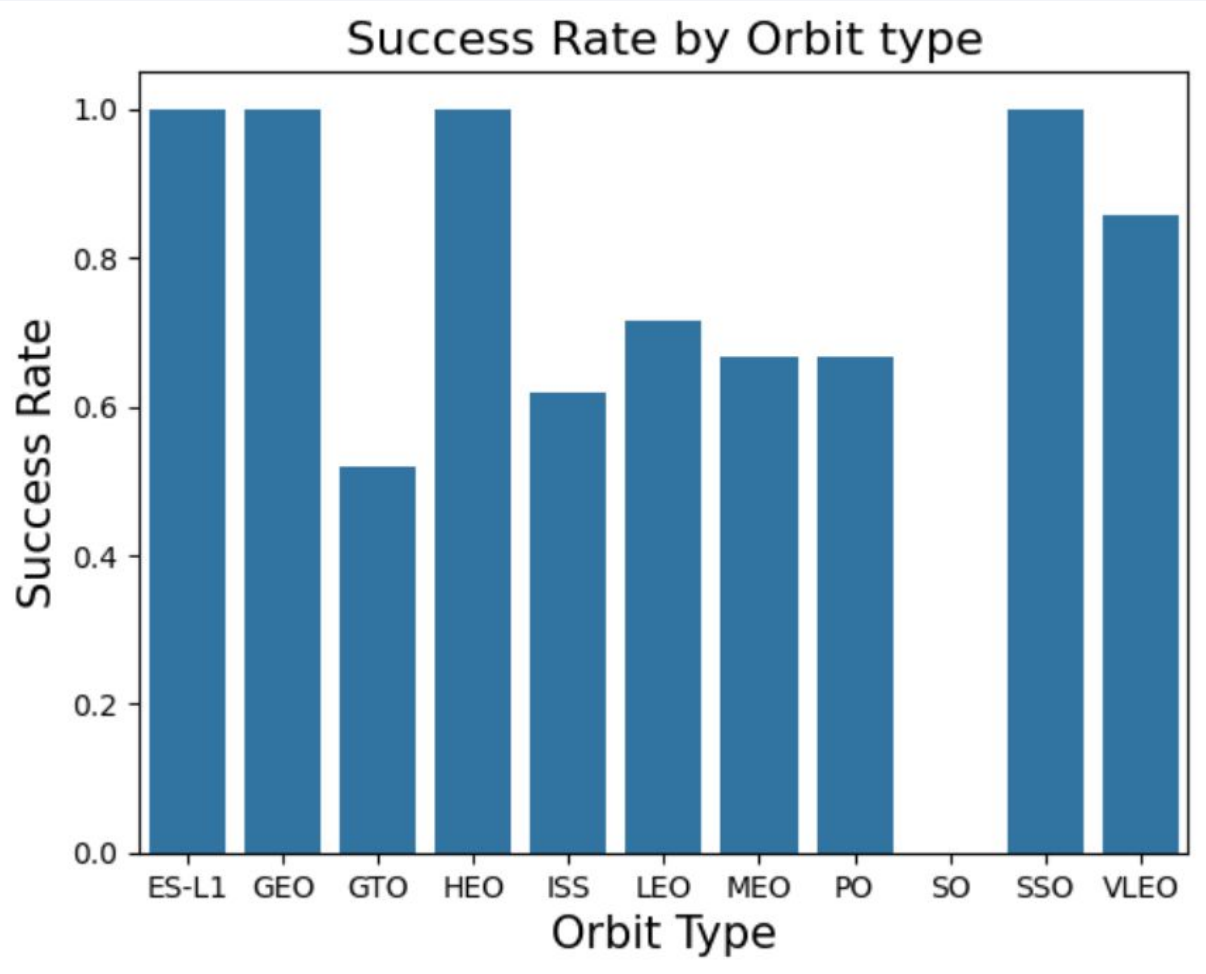
The flight number increases with respect to launch outcomes. Some launch sites have higher concentration of flights and these sites became more frequently used over time. Early flight numbers show more failures than later ones.

Payload vs. Launch Site



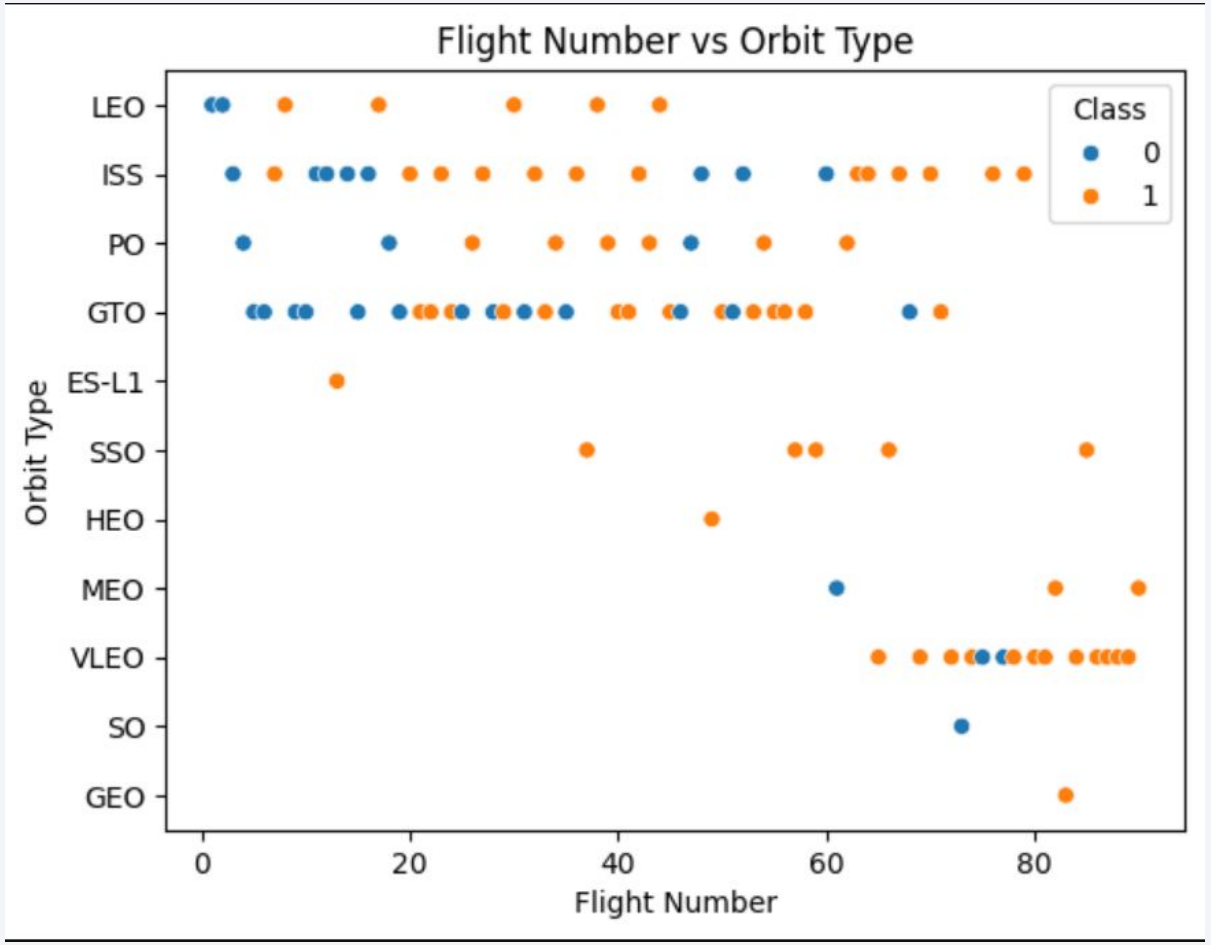
The plot indicates that landing success varies across launch sites and payload masses. Sites such as **KSC LC-39A** and **CCAFS SLC-40** show a higher number of successful landings, suggesting that launch conditions and payload mass influence Falcon 9 landing outcomes.

Success Rate vs. Orbit Type



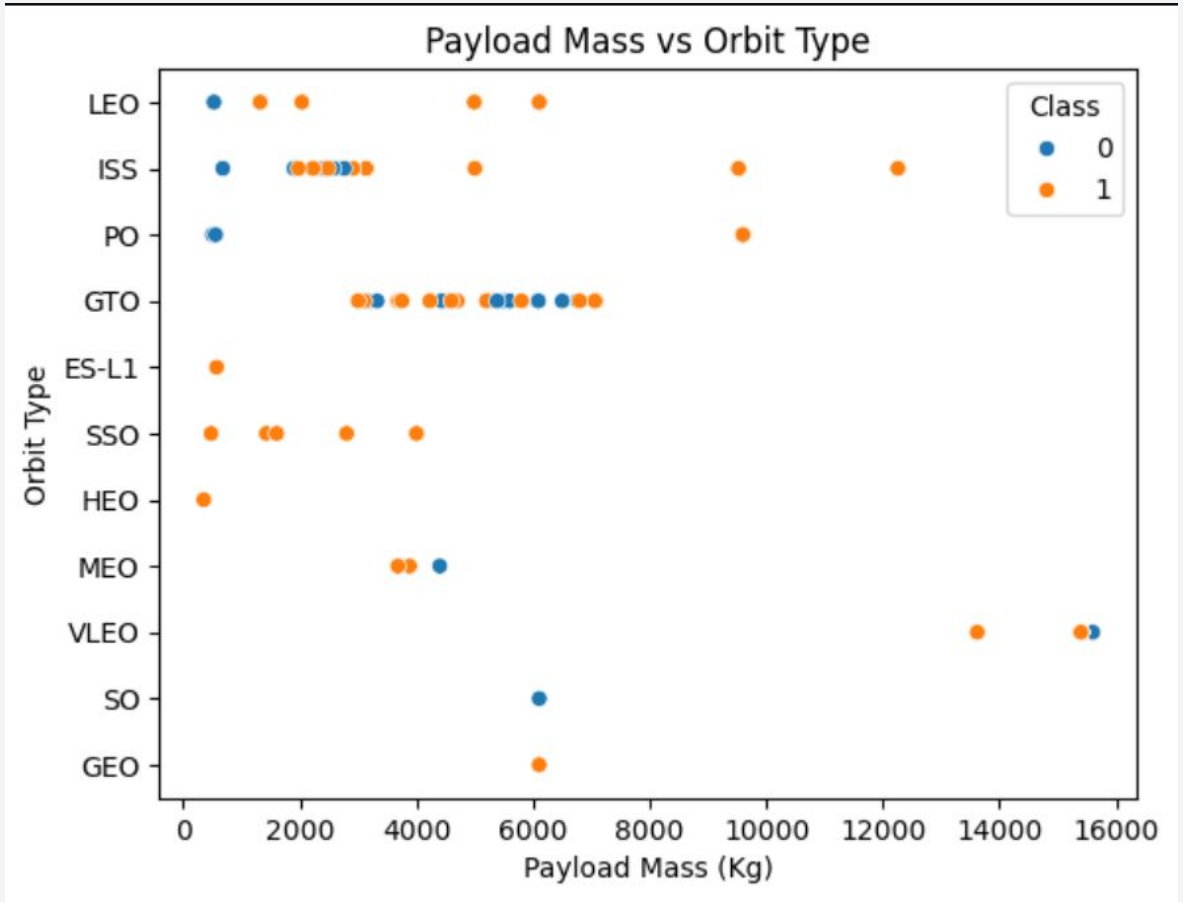
The success rate of Falcon 9 landings varies depending on the orbit type. Some orbits such as **ES-L1, GEO, HEO, and SSO** show very high success rates, while **GTO missions** have lower landing success. This indicates that **mission orbit requirements influence the probability of a successful landing.**

Flight Number vs. Orbit Type



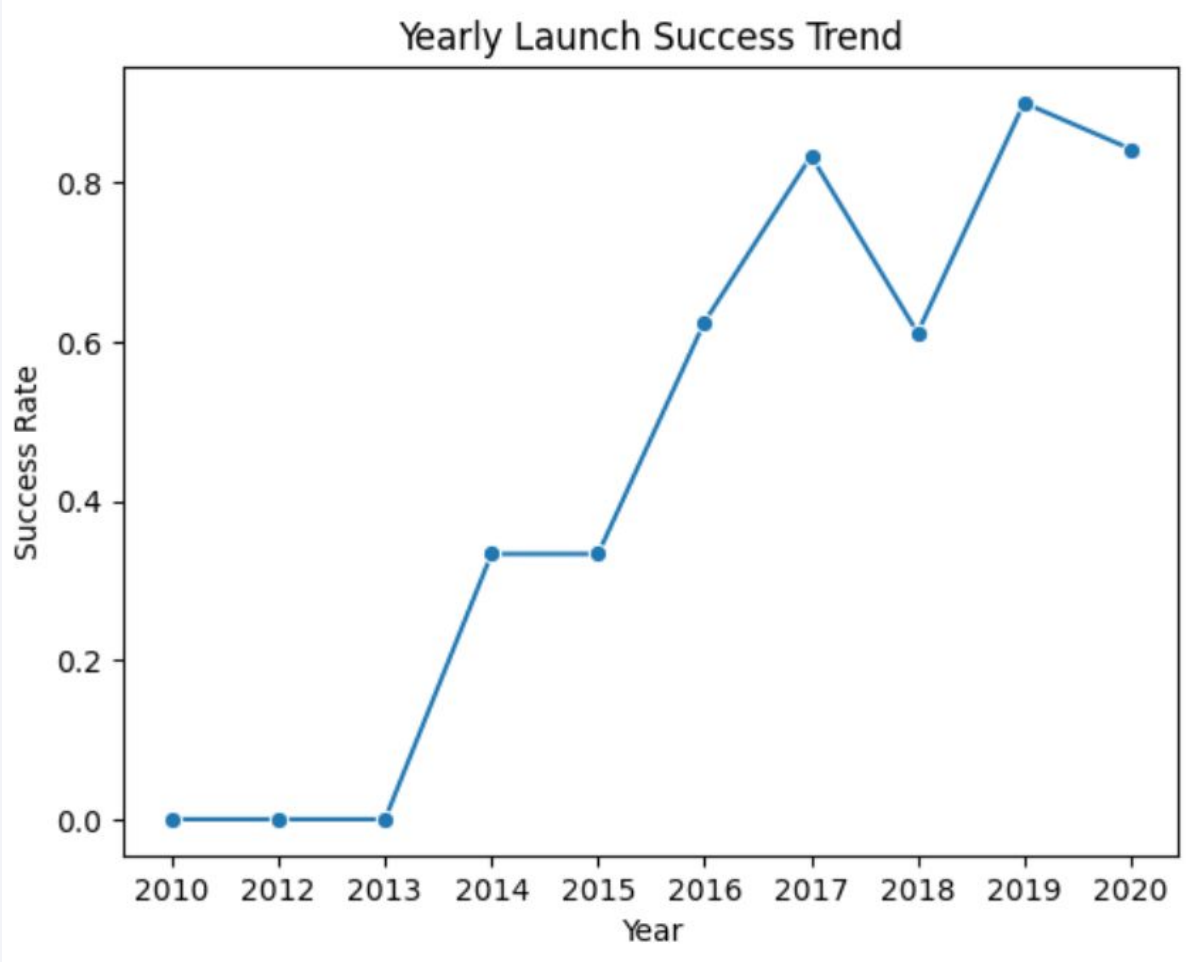
The plot shows that **landing success increases with flight number**. Earlier missions experienced more failures, while later launches achieved more successful landings, indicating **technological improvements and operational learning over time.**

Payload vs. Orbit Type



The plot shows that landing success varies depending on **payload mass and orbit type**. Missions to **LEO and ISS** generally achieve higher success rates, while **GTO missions** display a mix of successes and failures, suggesting that mission requirements affect landing outcomes.

Launch Success Yearly Trend



The success rate of Falcon 9 landings has **significantly improved over time**. Early missions had low success rates, but from **2016 onward**, the rate increased to around **80–90%**, highlighting continuous technological improvements and operational learning at SpaceX.

All Launch Site Names

Launch Site	Location	Description
CCAFS LC-40	Cape Canaveral, Florida	Falcon 9 orbital launch pad
VAFB SLC-4E	Vandenberg, California	Launch and landing site for Falcon 9
KSC LC-39A	Kennedy Space Center, Florida	Historic launch pad used by SpaceX

Launch Site Names Begin with 'CCA'

FlightNumber	PayloadMass	Orbit	LaunchSite	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
1	6104.959411764706	LEO	CCAFS SLC 40	1	False	False	False	nan	1.0	0	B0003
2	525.0	LEO	CCAFS SLC 40	1	False	False	False	nan	1.0	0	B0005
3	677.0	ISS	CCAFS SLC 40	1	False	False	False	nan	1.0	0	B0007
4	500.0	PO	VAFB SLC 4E	1	False	False	False	nan	1.0	0	B1003
5	3170.0	GTO	CCAFS SLC 40	1	False	False	False	nan	1.0	0	B1004

Total Payload Mass

SUM("Payload_Mass__kg_")
45596

45 596 kg is the total payload mass transported by the launches included in the query.

Average Payload Mass by F9 v1.1

AVG("Payload_Mass__kg_")
2928.4

If the query filtered launches for a **specific booster version (e.g., Falcon 9 v1.1)**, then **2928.4 kg** represents the **average payload mass delivered by that booster across its missions**.

First Successful Ground Landing Date

first_groundpad_success_Date
2015-12-22

December 22, 2015 marks the first successful Falcon 9 ground landing, a key milestone in reusable rocket technology.

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version
F9 FT B1029.1
F9 FT B1036.1
F9 B4 B1041.1

Three boosters (**F9 FT B1029.1, F9 FT B1036.1, F9 B4 B1041.1**) successfully landed on a **drone ship** while carrying payloads between **4000 kg and 6000 kg**.

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Most missions were **successful (98 launches)**, with only **one failure in flight** and a few other special success cases.

Boosters Carried Maximum Payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Several **Falcon 9 Block 5 boosters** carried the **maximum payload mass**, showing that this version supports the heaviest payload missions.

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

In **2015**, two missions had **drone ship landing failures**, both launched from **CCAFS LC-40** using **Falcon 9 v1.1 boosters**.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

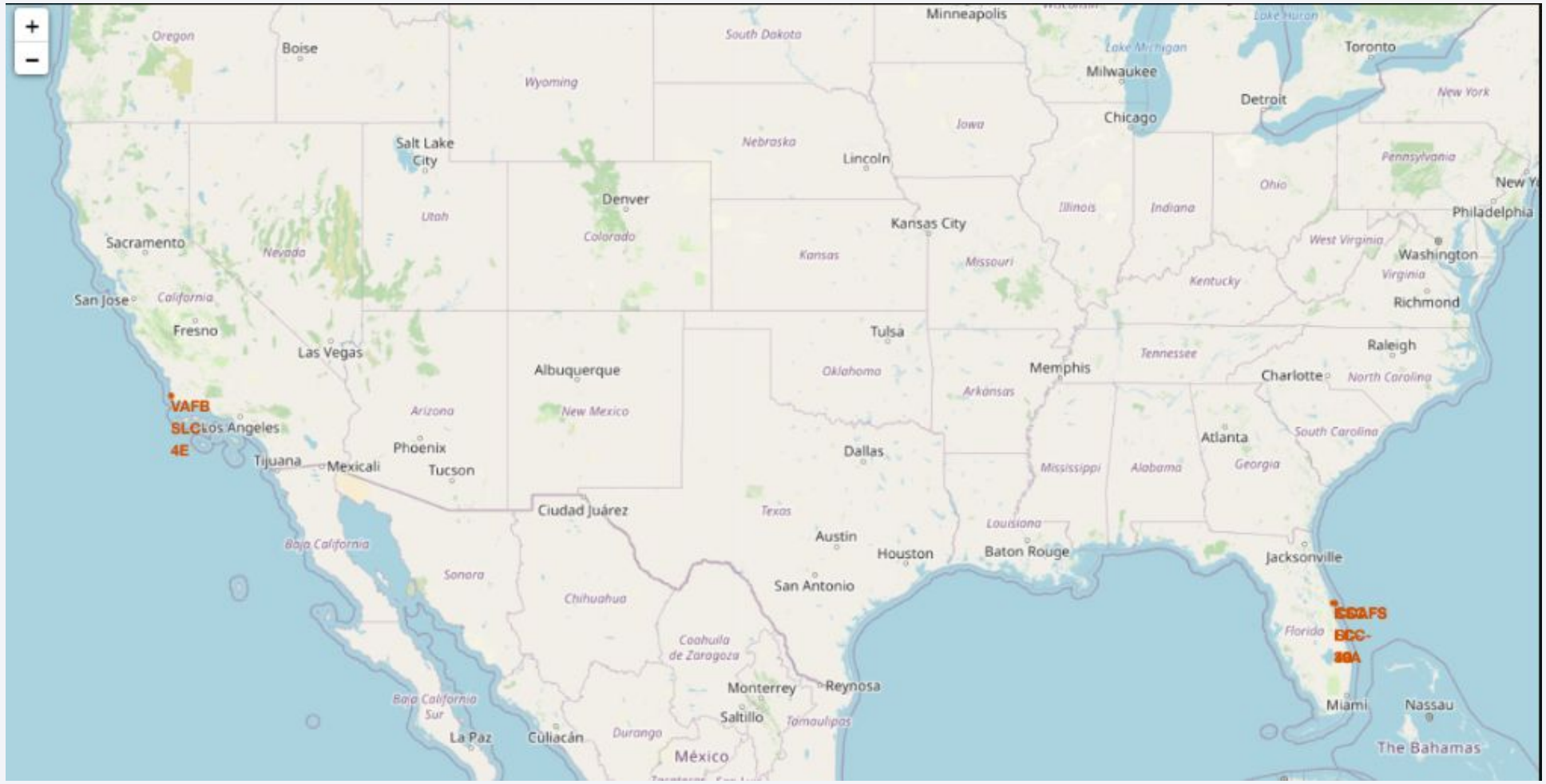
Landing_Outcome	Total
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Between **2010-06-04 and 2017-03-20**, the most frequent landing outcome was **No attempt (10 launches)**, followed by **Success (drone ship)** and **Failure (drone ship)** with **5 launches each**.

Section 3

Launch Sites Proximities Analysis

Launch Sites



Distance Between Launch Site and Coastline

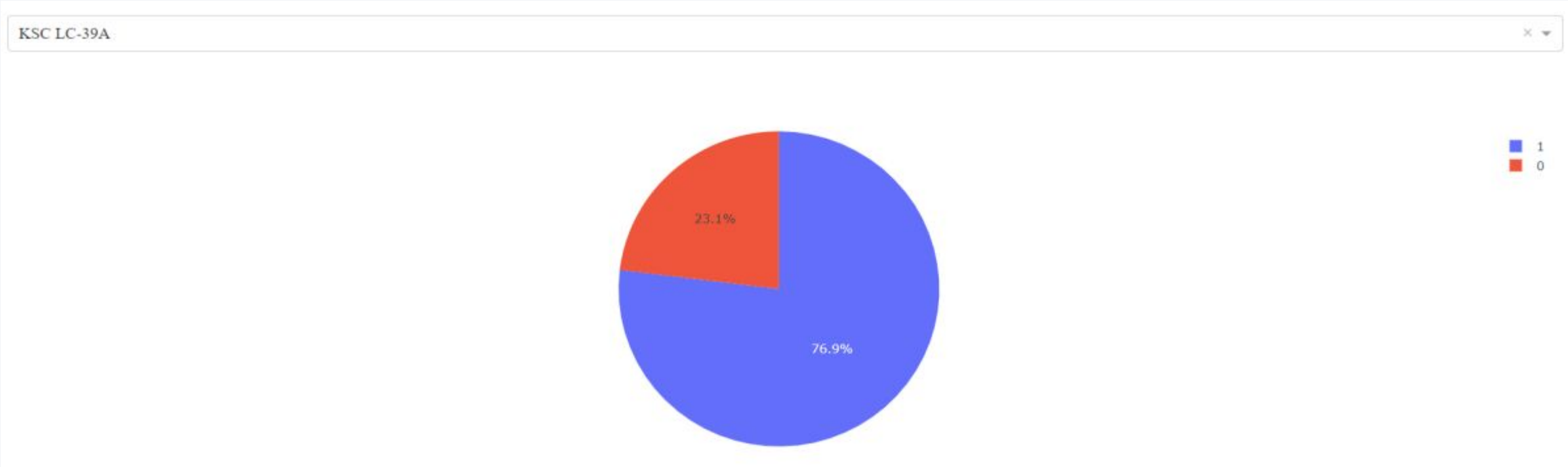
Section 4

Build a Dashboard with Plotly Dash

Total Sucessfull Launches By Site

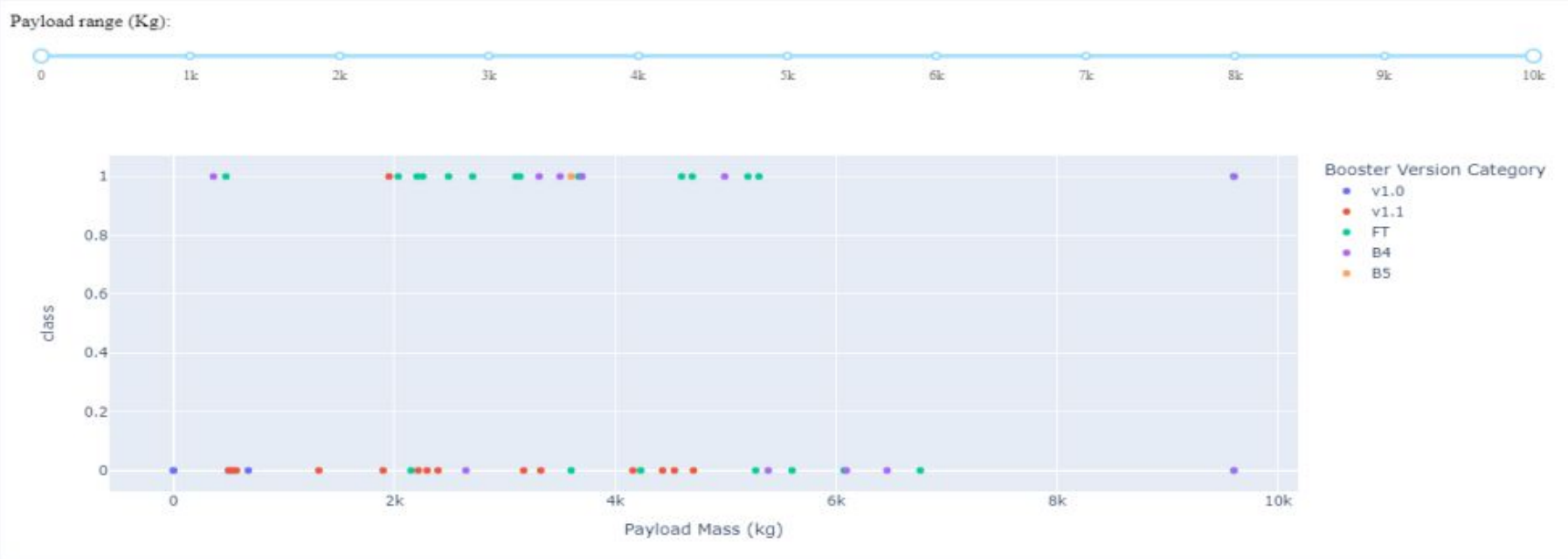
Most launches were conducted from **KSC LC-39A (41.7%)**, followed by **CCAFS LC-40 (29.2%)**, **VAFB SLC-4E (16.7%)**, and **CCAFS SLC-40 (12.5%)**, indicating that **KSC LC-39A is the primary launch site**.

Pie chart of KSC LC 39A



At **KSC LC-39A**, the majority of launches were **successful (76.9%)**, while **23.1% resulted in failure**, showing a **high landing success rate at this launch site**.

Payload vs. Launch Outcome scatter

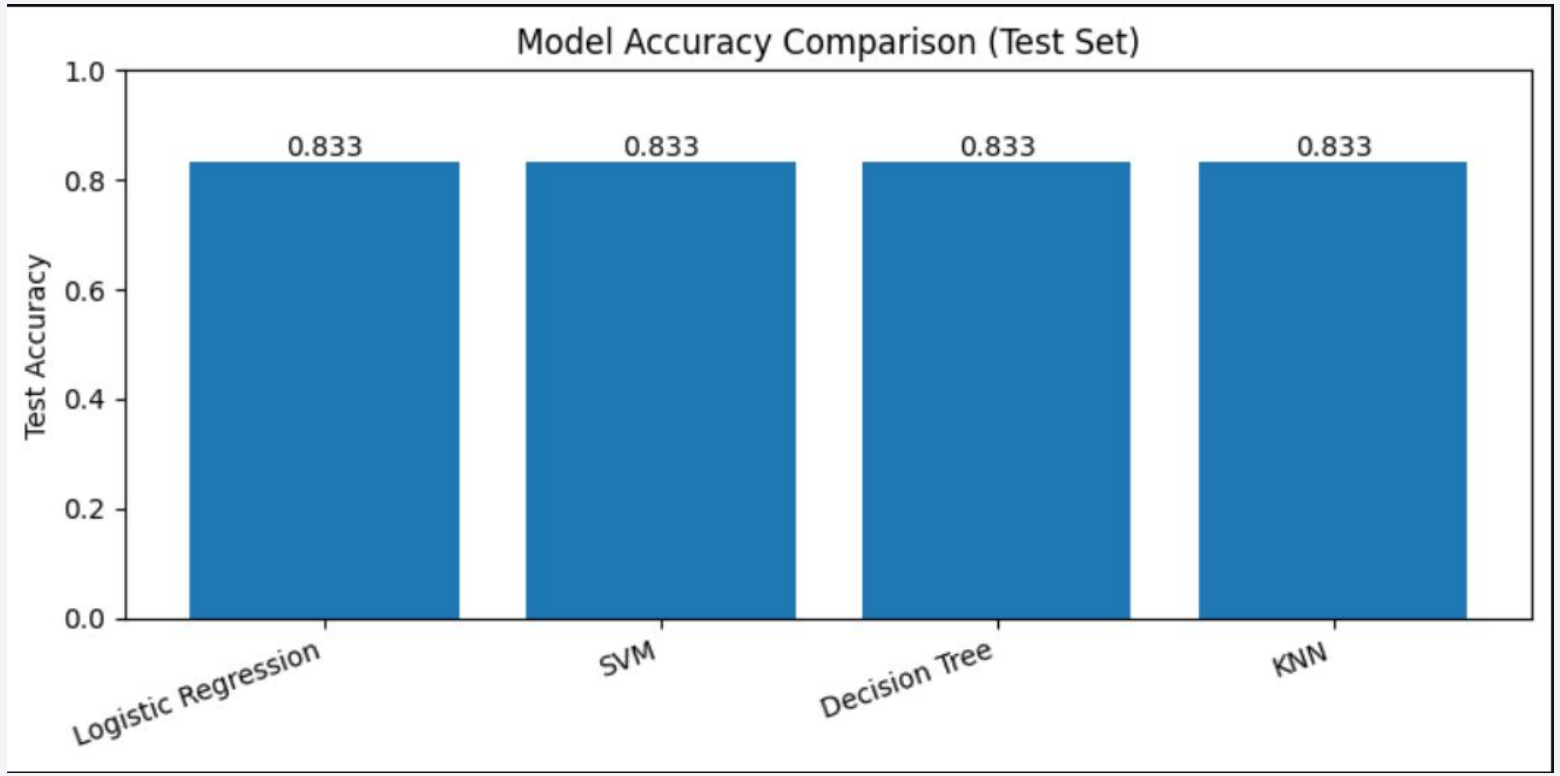


The scatter plot shows the relationship between **payload mass and landing outcome**, colored by **booster version category**. Successful landings (class = 1) occur across different payload ranges, but they become more frequent with **newer booster versions such as FT, B4, and B5**. Earlier versions like **v1.0 and v1.1** show more failures. This suggests that **improvements in booster technology increased landing success over time**.

Section 5

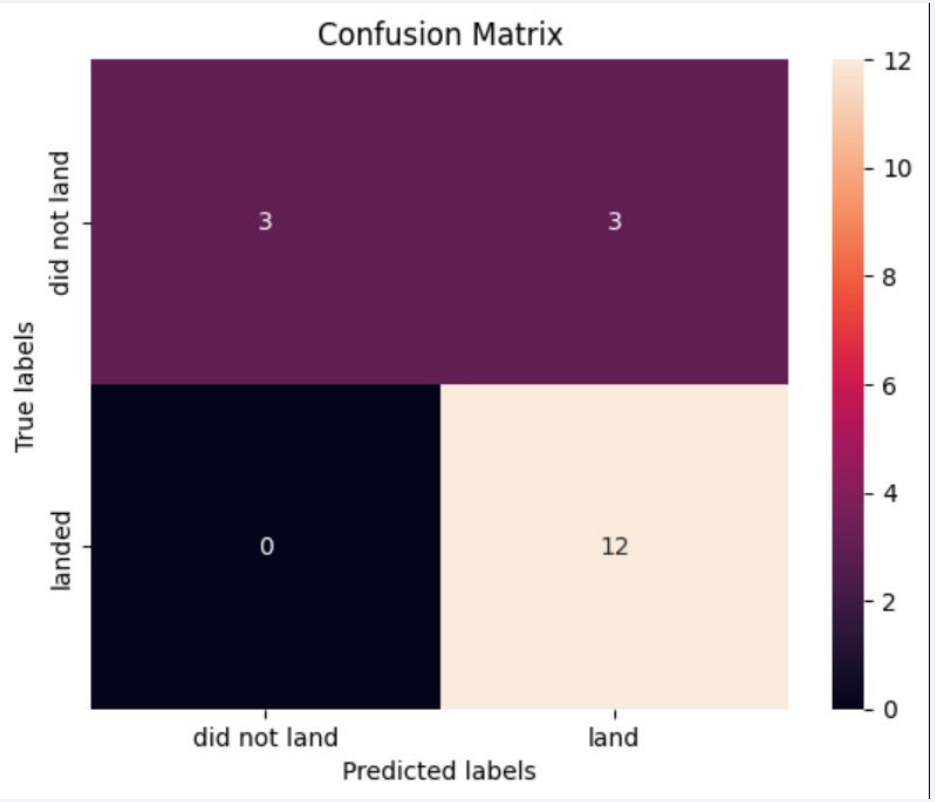
Predictive Analysis (Classification)

Classification Accuracy



The bar chart compares the **test accuracy of four classification models**: Logistic Regression, SVM, Decision Tree, and KNN. All models achieved a similar performance with an **accuracy of approximately 0.833 (83.3%)** on the test dataset. This indicates that each model is capable of predicting Falcon 9 landing success with comparable effectiveness, and no single model significantly outperforms the others.

Confusion Matrix



The confusion matrix shows the same model's classification performance for predicting Falcon 9 landing outcomes. The model correctly predicted **12 successful landings** and **3 failed landings**. However, **3 failed landings were incorrectly predicted as successful**, while there were **no cases where a successful landing was predicted as a failure**. Overall, all model demonstrates good performance in identifying successful landings but shows some difficulty distinguishing certain failure cases.

Conclusions

- Data was collected from the **SpaceX public API**, converted from JSON to a pandas DataFrame, and cleaned to focus on **Falcon 9 launches**.
- Exploratory Data Analysis identified important factors affecting landing success such as **payload mass, orbit type, launch site, and booster version**.
- The dataset was prepared using **feature scaling and a train–test split**, and several models were trained: **Logistic Regression, SVM, Decision Tree, and KNN**.
- After **hyperparameter tuning with GridSearchCV**, all models achieved similar performance with an **accuracy of about 83% on the test set**.
- Overall, the results show that **machine learning can effectively predict Falcon 9 landing outcomes**, highlighting the potential of data-driven methods for reusable rocket mission analysis.

Appendix

Additional materials and project resources are available in the GitHub repository.

GitHub Repository:
https://github.com/o-phys/spacex-falcon9-landing-prediction/tree/main

The repository includes:

- Data collection and preprocessing notebooks
- Exploratory Data Analysis (EDA)
- Interactive launch site visualizations
- Machine learning models used to predict Falcon 9 landing success

All notebooks and code used in this project can be accessed and reproduced through the repository.

Thank you!